

Ensuring Security on Open Source Virtual Platforms

We must thank IT virtualisation, as it led us to cloud technology. Today's IT infrastructures are already running their mission-critical business applications on virtual machines. Like physical infrastructure, virtualisation is also cursed with cyber security challenges. This article talks about a typical open source virtualisation solution, along with the steps to secure it.



As we know, a virtual server is a resource-based instance of an operating system. It consumes the pre-configured amount of CPU speed, memory and disk space from the physical hardware. An important low-level software component called HAL (Hardware Abstraction Layer) sits in between the physical hardware and the operating system (OS). HAL is responsible for converting OS calls into a set of CPU instructions and vice-versa, thus enabling high-level application stacks to run smoothly. HAL also makes the OS independent of the hardware components, and thus makes it portable across multiple physical platforms.

Virtualisation is achieved by adding another low-level software layer along with HAL, which allows us to simulate multiple virtual machines on a single physical

machine, whereby each virtual machine (VM) can run its own (copy of an) OS. Since this layer works together with the HAL, it provides great flexibility for us to select the OS, or assign a dedicated CPU and other resources to a particular machine. Figure 1 shows the basic building blocks of the virtualisation architecture. A component called the virtual session manager runs in parallel with guest instances in order to allocate system resources and monitor usage. In theory, there are three types of virtualisation. The first type is called a hypervisor, which means it is fully virtual, wherein all guest instances are kept separate from each other. In the second type, called para-virtualisation, the guest systems know about each other. This is achieved by semi-abstracting of the physical